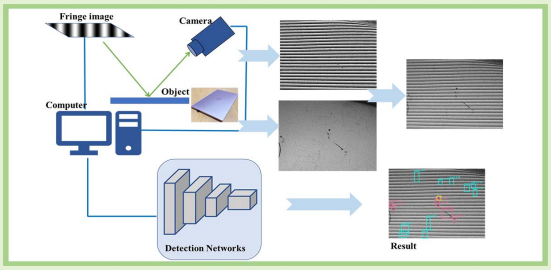
Smooth Surface Defect Detection by Deep Learning Based on Wrapped Phase Map

Jiamin Tao, Yongjian Zhu, Wenyi Liu, Frank Jiang, and Hongzhan Liu

***Abstract*—Object detection is widely used in the industry for visual inspection of defects. But it fails to fully reflect the defects on the smooth surface due to the difficulty in illuminating, and the high performance of defect detection tends to be a challenge. In this paper, we proposed a novel defect detection method for laptop panels. On the one hand, we utilized the wrapped phase map obtained by phase-shift reflection fringes for defect detection and used the DenseFuse network to fuse the wrapped phase map with the fringe modulation map to optimize the dataset. On the other hand, we proposed an improved deep learning-based network Vovecwnet. The experiments indicated that our image-based multi-information fusion system could achieve excellent performance in defect detection on the smooth surface (e.g., laptop panels surface), with high accuracy of 97.5% and a few reference time 0.0762s on GTX 2080ti GPU. Thus, our proposed method outperformed and tackled the challenges encountered in the defect detection of smooth surfaces.**



***Index Terms*— Surface defect detection, wrapped phase, multi-information fusion, deep learning.**

I. Introduction

DEFECT detection plays a vital role in the quality control of products. Each laptop panel has to go through multiple processes and cross-checks before it enters the market. Human vision-based defect detection methods are obsolete, while the automatic inspection of defects is a significant trend in the industry. It is necessary to implement further research on enhancing the efficiency and reliability of machine vision-based defect detection methods [1]–[3].

Machine learning-based defect detection methods have been widely adopted in industrial production, and some achievements have been obtained [2]. Traditionally, some hardware, such as an optical light source and mechanical control unit, is used to collect images. Image processing is performed through edge detection [4] and sub-pixel analysis, and finally, the data is analyzed. In this process, features of different defects are extracted manually. Additional rules are set to build models, and machine learning algorithms are used to determine whether there are defects by limiting their size or circumference [5]. Recently, the successful development of artificial intelligence, especially deep learning, has brought new concepts and ideas to machine vision. The emergence of multiple State-Of-The-Art (SOTA) detectors such as Faster-RCNN [6], SSD [7], YOLO [8], [9] provides unique solutions for defect detection in the industry. Currently, the object detection algorithm has been applied to defect detection in several studies. For example, Lei *et al.* [10] proposed a defect detection network for polymeric polarisers based on Faster-RCNN. The SSD-CNN model was adopted to detect roller surfaces, and good results were achieved [11]. In addition, Luo *et al.* [12] proposed a lightweight CNN-based network for cell phone surface scratch detection with an accuracy of 96.35%.

The difficulties in detecting appearance defects mainly come from the detected target objects. Defect features are most visible (e.g., texture, colours, shapes) with a little semantic information, making it difficult to set strict rules to determine thresholds. As a result, the utilization of defect detection methods based on traditional machine learning methods is

Manuscript received April 9, 2021; accepted April 26, 2021. Date of publication April 29, 2021; date of current version July 14, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 61875057, Grant 61475049, and Grant 61774062; in part by the Science and Technology Program of Guangzhou under Grant 2019050001; and in part by the Public Projects of Zhejiang Province under Grant LGG21E050006. The associate editor coordinating the review of this article and approving it for publication was Dr. Amitava Chatterjee. *(Corresponding authors: Yongjian Zhu; Hongzhan Liu.)*

Jiamin Tao, Wenyi Liu, and Hongzhan Liu are with the Guangdong Provincial Key Laboratory of Nanophotonic Functional Materials and Devices, School for Information and Optoelectronic Science and Engineering, South China Normal University, Guangzhou 510006, China (e-mail: lhzscnu@163.com).

Yongjian Zhu is with the School of Mechanical and Energy Engineering, Zhejiang University of Science and Technology, Hangzhou 310023, China, and also with Ningbo Mizhong Information Technology Company Ltd., Cixi 315302, China (e-mail: zhuyongjian_hn@126.com).

Frank Jiang is with the Faculty of Engineering and IT, University of Technology Sydney, Sydney, NSW 2007, Australia (e-mail: frank.jiang@uts.edu.au).

Digital Object Identifier 10.1109/JSEN.2021.3076610

1558-1748 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 07:17:48 UTC from IEEE Xplore. Restrictions apply.

limited [2]. Although deep learning can solve some problems that conventional machine vision techniques cannot solve, it also causes other issues. In fact, common object detection tasks are almost high-level computer vision tasks, in which a small proportion of visual features is used. The features of the detected objects contain explicit semantic relationships, and models are used to extract semantic information [6]–[9]. However, defect detection is different from other object detection tasks. For example, the defects of laptop panels mainly contain visual features. Still, they lack semantic information such as partial and overall relations, leading to relatively poor performance when using traditional object detection methods. Consequently, some appropriate modifications to conventional object detection methods are required for defect detection tasks. Besides, to achieve satisfactory detection results, apart from the excellent algorithm, suitable image acquisition methods are needed to meet the requirements of different defects [13]. The detection of light-reflecting objects commonly requires a complicated illumination system, such as the bright-dark field [14], backlight [15], and coaxial light sources [16]. In general, these illuminations lack robustness and should be carefully selected and set up; otherwise, it would generate a large area of white spots on images so that the defect features are hard to capture. Another option is to employ the fringe projection technique with a flat fringe light source [17], [18]. The fringe patterns are projected onto the object, which will be deformed when the surface of the object is uneven, and minor defects on the surface of products can be detected. The use of fringe projection techniques makes up for the shortcomings of the above-mentioned light sources, but it also fails to fully reflect the defects of smooth surfaces. In fact, the Illumination problem is an essential task of computer vision. Some approaches for invariant feature extraction can help in solving the problems. The histogram equalization method was proposed to extract features independent of imaging conditions [40]. The von Kries model is also used to estimate the images that are sensitive to the acquisition device and illuminant change [41]. However, more recent works focused on colour feature variation, which is a challenge for dealing with such a complex task as defect detection.

To overcome the above difficulties, we present an improved defect detection method of reflective surfaces with three strategies, (a) phase-shift fringe reflection method [19] to record the wrapped phase distribution [20], [21]and the fringe modulation [19] of the defects on the laptop panels surfaces, (b) the multi-information fusion of wrapped phase map and fringe modulation map based on the image fusion network DenseFuse [22] to optimize the dataset, and (c) a new backbone VoVecwNet, extends VoVNetV2 [23], with efficient extraction capabilities for defect features. The principles are based on an analysis of the fringe phase information, which allows surface imperfection caused by defects to be identified by the models easily. In this way, the complex process of lighting will be greatly avoided and defects that would otherwise remain undiscovered can be spotted. Compared to the above-mentioned techniques, our method improves the performance of defect detection tasks.

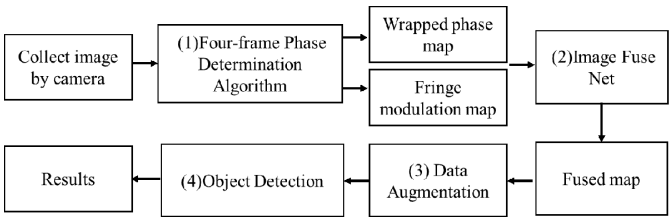


Fig. 1. Process of defect detection for laptop panel surface.

II. Related Work

Significant progress has been made in defect detection due to the development of deep learning. Neural networks are employed for defect detection of products surface in several papers. Most recently, Lin *et al.* [24] completed the detection of a casting defect tracked by a deep convolution neural network. Likewise, the multitask networks have been explored for wire defect recognition of spring-wire socket [25]. Additionally, Mobile Net-SSD was applied first in a surface defect detection in [26], driven by a lightweight network. Chen *et al.* [27] utilized a deep learning architecture to inspect the surface scratches of the plastic housing optically. Huang *et al.* [28] presented an MCuePush Unet model for the saliency detection of magnetic tile surface defects.

The above-mentioned approaches mainly focus on improving the algorithm while ignoring the fact that the effect of detection is also heavily dependent on the quality of the data sent to the models. They are limited to obtaining useful information from conventional optical images for defect detection, requiring a complicated lighting system. To simplify the lighting system and consider that the defects on the product surface are often visual features with weak semantic information, we creatively used multi-information fusion maps rather than conventional optical images to finish defect detection tasks.

III. Proposed Method

In this paper, a defect detection method of the smooth surfaces is proposed, involving a novel backbone and multi-information fusion mechanism for defect detection on laptop panels. Fig. 1 illustrates the process of detection. Firstly, the collected fringe image is processed by an algorithm to obtain the wrapped phase and fringe modulation map, with 1080×1000 pixels in resolution. Secondly, The image fusion net is adopted to fuse the wrapped phase map and fringe modulation map to obtain the fused map. Thirdly, several data augmentation methods are applied to expand data samples, involving random flipping, cropping, rotation, and filtering. Finally, the images which have been labelled manually are fed into the object detection network for training. Then the trained object detection model is used to detect and classify defects.

A. Image Acquisition

The choice of the light source is crucial to accurately detect defects in the detection system based on machine vision. The different effects of the same defect on a smooth surface can be obtained under various light sources. It can

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 07:17:48 UTC from IEEE Xplore. Restrictions apply.

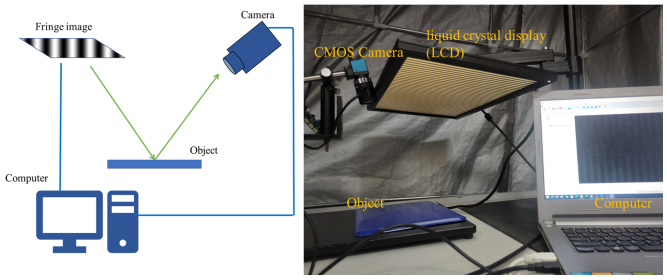


Fig. 2. Hardware structure based on our fringe method.

be challenging to make the right decision even for the professional staff. A standard method in dealing with specular reflection is the use of bright-dark field illumination in industrial machine vision inspection, which increases image-processing complexity. We adopted the fringe reflection method for image acquisition to overcome these problems. The whole system involves hardware and software. As shown in Fig. 2, the system's hardware comprises a sinusoidal fringe pattern generator (1900*600 liquid crystal display (LCD)), CMOS industrial camera, and computer. The software consists of four major stages, image acquisition unit, image processing unit, image fusion model, and object detection model.

In this process, the surface of the detected object reflects the fringe patterns generated by the LCD computer. Then the reflected fringe patterns are captured by the CMOS industrial camera. The defects on the surface of the object lead to the deformation of the fringe patterns. These deformed fringes contain the defect information on the surface. The corresponding algorithm is used to process the deformed fringe to obtain the wrapped phase [20], [21] and fringe modulation [19] that contains the surface topography information. The use of wrapped phase maps instead of traditional optical images for object detection has two advantages, i.e., the phase map can provide more semantic information for subsequent network learning. Moreover, it addressed the lighting challenges for image acquisition of highly smooth objects.

The phase-shifting method [20] is applied to analyze the sinusoidal pattern reflected from the object and obtain the wrapped phase and fringe modulation. Four-step sinusoidal fringe patterns with a phase shift of $\pi/2$ are projected on the laptop panel, which can be summarized as:

$$I_1(x, y) = A(x, y) + B(x, y)\cos[\emptyset(x, y)] \quad (1)$$

$$I_2(x, y) = A(x, y) + B(x, y)\cos[\emptyset(x, y) + \pi/2] \quad (2)$$

$$I_3(x, y) = A(x, y) + B(x, y)\cos[\emptyset(x, y) + \pi] \quad (3)$$

$$I_4(x, y) = A(x, y) + B(x, y)\cos[\emptyset(x, y) + 3\pi/2] \quad (4)$$

where $A(x, y)$ represents background intensity, $B(x, y)$ represents fringe contrast, and $I(x, y)$ represents the intensity of fringe patterns. They are all given at pixel.

The wrapped phase distribution $\emptyset(x, y)$ can be calculated according to (5).

$$\emptyset(x, y) = \arctan\left(\frac{I_2 - I_4}{I_1 - I_3}\right) \quad (5)$$

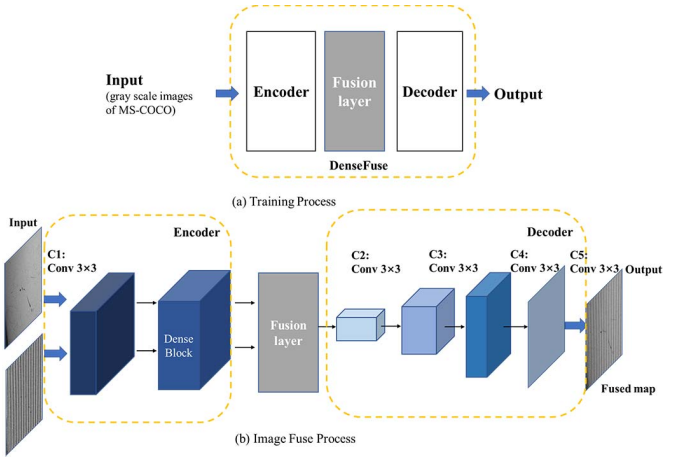


Fig. 3. The architecture of densefuse. (The image fusion network consists of three parts: encoder, fusion layer, and decoder. The encoder (C1 and denseblock) is used to extract depth features. The fusion layer is used to fuse features. Finally, the fusion image is reconstructed by the decoder.)

The fringe modulation $M(x, y)$ can be calculated according to (6).

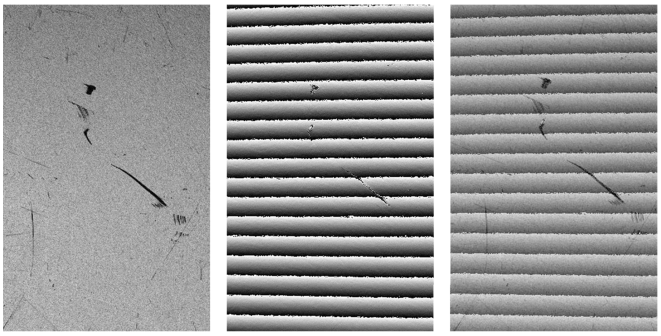
$$M(x, y) = 2 \times \frac{\sqrt{(I_1 - I_3)^2 + (I_2 - I_4)^2}}{I_1 + I_2 + I_3 + I_4} \quad (6)$$

B. Image Fusion

The images fed into the network are the fused maps which are the combination of wrapped phase and fringe modulation maps. They are both produced by the phase-shifting method mentioned above. The fused data retains the valuable information of two image sources as much as possible, which is a better representation of the defect characteristics. Through multi-information fusion, the detection accuracy can be further improved [2]. The use of the fused maps for the object detection model's training can enhance the robustness of the network. We used the same industrial camera to collect data to ensure strict alignment between the wrapped phase and the fringe modulation maps. The wrapped phase map and fringe modulation map are sent to DenseFuse [22], which has been trained on the grayscale images of MS-COCO [29] to obtain the fused map. The DenseFuse structure is depicted in Fig. 3.

The accuracy of defect recognition based on the wrapped phase maps is significantly higher than that of conventional optical images. The information is compressed in the process of extracting the wrapped phase distribution. As a result, when the neural network analyzes and processes the maps, the memory and time cost will be reduced. The calculation speed will be higher compared to the conventional optical images for object detection. Although part of the detailed features is lost when the wrapped phase maps are generated, as depicted in Fig. 4, at this time, the modulation maps can play a role in making up for the loss of features. DenseFuse utilizes the temporal and spatial correlation and information complementarity of two (or more) images, allowing the images obtained by fusion to reflect a more comprehensive description of defects than before, which is more conducive to the detection task. The subsequent experiments also proved that

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 07:17:48 UTC from IEEE Xplore. Restrictions apply.



(a) Fringe Modulation Map (b) Wrapped Phase Map (c) Fused Map

Fig. 4. Three types of defect maps on the laptop panel.

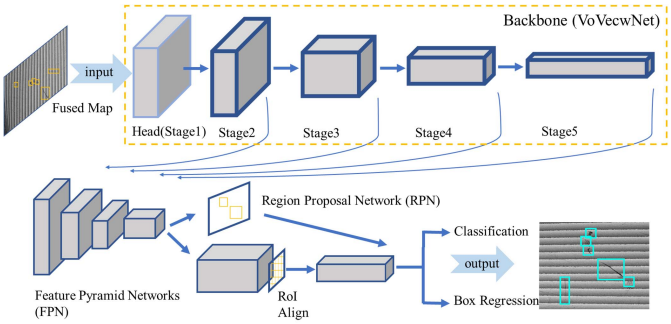


Fig. 5. The architecture of the detection network.

image fusion has a good expression of defects, and the multi-information fusion has an unbeatable advantage.

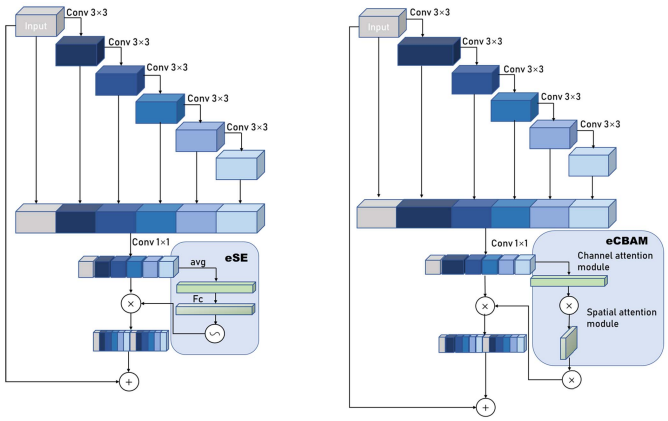
C. Detection Network

The detailed detection network structure based on the fused maps is depicted in Fig. 5. Defect detection is mainly to process visual features that require a network with fewer layers to extract. Taking this into account, we propose VoVecwNet to serve as the feature extraction part of the detection networks. VoVecwNet is based on VoVNetV2 [22], an improved version of DenseNet [30]. In VoVNetV2, intermediate feature maps with multi-receptive fields are aggregated on the final layer of each module. Moreover, the shortcut connection of ResNet [31] and SE module (Squeeze-and-Excitation attention block) [32] is added. VoVNetV2 increases the proportion of the visual features in the final decision for classification tasks suitable for defect detection. To achieve better performance in defect detection, we propose VoVecwNet based on VoVNetV2. This proposed backbone is composed of stacking in five stages. Each basic module is shown in Fig. 6. Details about the VoVecwNet structure can be seen in Table I.

There are two refinements to boost the performance of VoVecwNet in defect detection. (a) The channel size of 3×3 convolution is increased in the first module. The 3×3 convolution plays a crucial role in the network, which is the only component that can facilitate the learning of feature maps [33]. However, in the original VoVNetV2, the number of channels received by the shallow layers is small, limiting the learning capability of the networks. It requires a maximum channel to enhance the ability to learn the visual features of the defects.

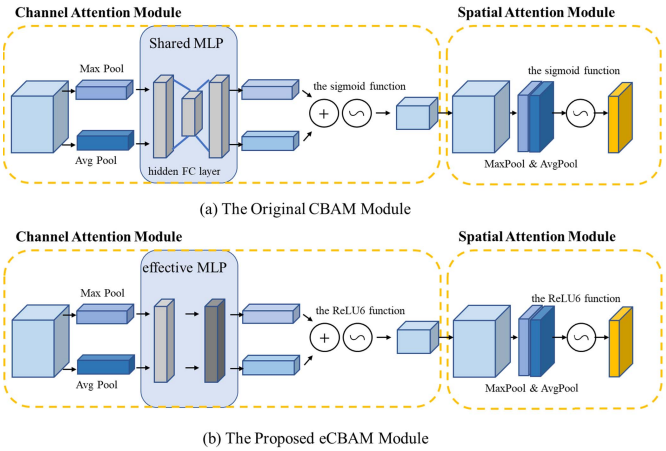
TABLE I
THE STRUCTURAL CONFIGURATION OF VOVECWNET

Type	VoVecwNet
head stage 1	3 × 3 Conv, 64, stride=2 3 × 3 Conv, 64, stride=1 3 × 3 Conv, 128, stride=1
stage2	module×1
stage3	module×1
stage4	module×4
stage5	module×3



(a) A Basic Module of VoVNetV2 (b) The First Module of VoVecwNet

Fig. 6. A basic module of VoVNetV2 and VoVecwNet.



(a) The Original CBAM Module

(b) The Proposed eCBAM Module

Fig. 7. CBAM module and eCBMA module.

(b) The attention mechanism is used in both dimensions of space and channel, which results in more vital representation ability than only used in one of them independently [34]. To enable the network to focus on these useful features properly, we used eCBAM (effective Convolutional Block Attention Module) instead of eSE (effective Squeeze-and-Excitation attention block) [23] in the backbone framework, as illustrated in Fig. 6. The original CBAM contains a hidden FC layer in a shared MLP block, as described in Fig. 7. The primary function of this layer is to squeeze the channel

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 07:17:48 UTC from IEEE Xplore. Restrictions apply.

dimension of the feature maps, which will cause the loss of useful information to a certain extent [23]. Therefore, the CBAM module is redesigned as the eCBAM module. In the refinement, we removed the hidden FC layer to obtain better spatial attention. As a result, more meaningful features in two dimensions of space and channel can be exploited.

The subsequent network structure follows the framework of Faster R-CNN [6]. We employ the feature pyramid network [35] to perform feature extraction on VoVecwNet and obtain feature maps of multiple scales. These feature maps are marked as fpn1, fpn2, fpn3, and fpn4. The feature maps at each scale, preserving information from various convolutional layers, which can effectively avoid the problem of information loss caused by the downsampling process. The RoIAlign, a new ROI assignment function proposed by Mask R-CNN [36], replaces the original Faster R-CNN's ROI pooling.

IV. Experiments

Techniques of our proposed method can be split into three parts, VoVecwNet backbone, wrapped phase map, and image fusion. In this section, we evaluate the effectiveness of our method thoroughly. Firstly, a series of ablation experiments are conducted. Then comparisons are performed between the proposed method and mainstream methods in the industry. The performance of models is reported in the same environment equipped with NVIDIA RTX 2080Ti GPU, CUDA v10.1, cuDNN v7.33, and PyTorch 1.3.

A. Description of Datasets

1) Description of NEU-DET: NEU-DET [37] is published by Northeastern University. The dataset includes 1800 grayscale images and contains six typical surface defects of hot-rolled steel. The defects are Rolled-in Scale (RS), Patches (Pa), and Crazing (Cr), Pitted Surface (PS), Inclusion (In), and Scratches (Sc). Each category contains 300 samples. For defect detection tasks, the dataset provides annotations to indicate the category and location of the defect in each image. Compared with other data sets, NEU-DET has less information. There is no unified standard to divide the boundary of defects, so it is a data set with a particular difficulty in detection.

2) Description of Our Dataset: Our dataset contains five types of images about surface defects of laptop panels, including conventional optical images, wrapped phase maps, fringe pattern maps, fringe modulation maps, and fused maps. These images were collected in the same scene using different image acquisition methods, with a total of 3,000 images. Each image had one or more defects. The defects were scratches, cracks, and spots. More details can be found in Fig. 8 and Table II. They were strictly annotated manually according to the ground truth provided by the laptop manufacturer.

B. Ablation Study

We conducted a series of ablation experiments to validate the efficiency of our three core techniques. The results are reported in mean Average Precision (mAP) to evaluate the performance of the network models.

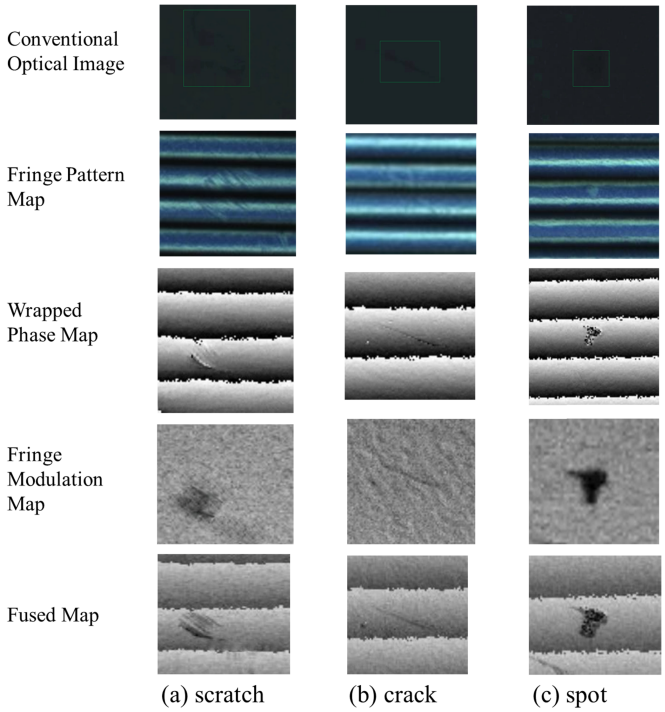


Fig. 8. Different imaging results of laptop panel defects; (1) The imaging effect of the dark field illumination (It can be seen that the panel surface is uniformly illuminated. The defects are in the green frame, but the defects are almost invisible.); (2) The imaging effect of the striped light source of Japan CCS company (The defects are more clearly visible.); (3) Wrapped phase map (fringe reflection method with the light source of the LCD) (The defect boundary is clear, and some details are missing.); (4) Fringe modulation map (fringe reflection method with the light source of the LCD) (blurry defect boundaries and rich details); (5) Fusion map (clear defect boundaries and rich details).

TABLE II
Details About Different Imaging

Detail	Image acquisition method	Light source
Conventional Optical Image	traditional method	dark field illumination
Fringe Pattern Image	fringe projection	flat fringe light source of Japan CCS company
Wrapped Phase Map	phase-shift fringe reflection	liquid crystal display (LCD)
Fringe Modulation Map	phase-shift fringe reflection	liquid crystal display (LCD)
Fused Map	phase-shift fringe reflection + image fusion	liquid crystal display (LCD)

1) The Proposed Backbone (VoVecwNet): To estimate the performance of the backbone we designed for defect detection on other object detection tasks, we compared the mAP of both the original and improved backbone on the public dataset PASCAL VOC 2007+2012 [38] and NEU-DET, which is the most widely used in defect detection. Pre-training is one common strategy to improve detection performance [39]. Therefore, to make a fair comparison, all the models were trained from scratch in this experiment, and the detector is Faster-RCNN.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 07:17:48 UTC from IEEE Xplore. Restrictions apply.

TABLE III
DETECTION RESULTS ON VOC 2007+2012 AND NEU-DET TEST SET

Backbone	Params	mAP(%) (on VOC)	mAP(%) (on NEU-DET)
VoVNetV2	58.99M	73.5	72.6
VoVNet+CBAM	53.88M	74.1(+0.6)	73.4(+0.8)
VoVNet+eCBAM	58.99M	74.4(+0.9)	74.4(+1.8)
VoVNet+CBAM+wise	55.97M	74.8(+1.3)	74.9(+2.3)
VoVNet+eCBAM+wise(ours)	61.07M	75.1(+1.6)	75.4(+2.8)

TABLE IV
COMPARISON WITH THE OTHER MODEL ON DEFECT DATASET (NEU-DET)

Model	FPS	mAP(%)
Faster-RCNN-r50	10	70.4
SSD300	34	68.4
YOLOv3	55	69.1
Ours	12	75.4

The experimental results can be found in Table III, which show that the improved model only adds a small number of parameters and calculations. Still, the detection performance is favourably improved compared with the original one, VoVNetV2, a backbone that outperforms ResNet and DenseNet with higher speed and better energy efficiency [23]. On the VOC dataset, by using eCBMA and adopting the strategy of increasing the underlying convolution kernel channel, the mAP has been improved to various degrees, which shows the effectiveness of our backbone refinement. The performance on the public dataset VOC is good, indicating that VoVecwNet can also serve as an efficient backbone for general detection. Simultaneously, the same experiment is conducted on the public defect dataset NEU-DET. It is found that the mAP of VoVecwNet boosts significantly, which demonstrates that VoVecwNet has a more remarkable ability to handle defect detection tasks.

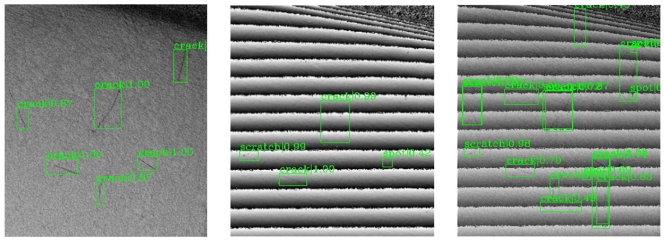
2) Comparison With the Other Model on Defect Dataset (NEU-DET): For further validation of VoVecwNet on defect detection, comparisons are performed between our VoVecwNet with other state-of-the-arts object detection models on the defect data set NEU-DET. The results are shown in Table IV.

The results indicate that VoVecwNet exhibits better performance than some other current mainstream networks on a dataset widely used in the defect detection field, which confirms the applicability of our model to defect detection tasks.

3) Image Fusion: To demonstrate the superiority of the image fusion method used in this paper in optimizing the dataset, we designed a comparison experiment on the dataset to test the effect of four different data types on the detection performance. Images are randomly divided into the training set, validation set, and test set according to 8:1:1. The training set is used for model training, the validation set is used to select model hyperparameters, and the test set is used to evaluate model performance.

TABLE V
DETECTION RESULTS ON DIFFERENT DATA

Detail	Conventional Optical Image	Fringe Pattern Image	Wrapped Phase Map	Fused Map
mAP(%)	73.5	87.9	88.4	93.5



(a) Detection Result on Fringe Modulation Map (b) Detection Result on Wrapped Phase Map (c) Detection Result on Fused Map

Fig. 9. Defect detection results on a laptop panel in the same scenario; (a) cracks are visible on fringe modulation map, and other types of defects are difficult to find; (b) scratch and spot defects are visible on the wrapped phase map, but cracks are basically invisible; (c) all the defects on the panel can be well detected on the fusion map.

TABLE VI
INFLUENCE OF DATA AUGMENTATION AND PRETRAINING ON THE NETWORK PERFORMANCE

Detection on fused map				
Data augmentation	No	Yes	No	Yes
Pretrained	No	No	Yes	Yes
mAP(%)	93.5	94.7	96.2	97.5

The experimental results are summarized in Table V. With the same algorithm, the selection of imaging methods determines the efficiency and accuracy of defect detection. Image fusion has a comprehensive description of the defects of the detected object. Therefore, the performance of the fused image is better than that of the wrapped phase map alone. The image fusion method boosts the robustness of the network and solves the drawbacks of the modulation and the wrapped phase data. The defect detection results on the laptop panels in the same scene are shown in Fig. 9. It isn't easy to set an appropriate illumination when getting conventional optical images of smooth objects. These images cannot fully reflect the defects in poor lighting conditions, resulting in poor detection performance. Our image acquisition methods have addressed the challenges of lighting. A problem that also arises in using the wrapped phase map is that it is easy to lose relatively shallow defects such as scratch features. At this time, the modulation map plays a role in making up for the loss of features. Therefore, in the comparative experiment, the performance of the fusion group is better than other image groups that have not been fused, which also illustrates the use of multi-information fusion is crucial to improve performance in defect detection.

Data augmentation and pre-training strategies affect model performance, and comparison results can be found in Table VI. A variety of data augmentation strategies are performed to improve our model's generalization and avoid overfitting,

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 07:17:48 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
DETECTION RESULTS ON DIFFERENT METHODS

Method	Scratch			Crack			Spot			Times(s)
	Recall	Precision	F1 Score	Recall	Precision	F1 Score	Recall	Precision	F1 Score	
Faster-RCNN-ResNet50	0.861	0.816	0.838	0.771	0.704	0.736	0.875	0.786	0.828	0.124
SSD300	0.754	0.658	0.703	0.572	0.328	0.417	0.526	0.489	0.507	0.067
YOLOv3	0.673	0.503	0.576	0.787	0.645	0.709	0.667	0.584	0.623	0.023
Ours	0.997	0.939	0.967	0.993	0.958	0.975	0.961	0.907	0.933	0.076

which involves horizontal flipping, rotation, cropping, low-pass, and high-pass filtering randomly while preprocessing the images. The best results are achieved when the model pre-trained on PASCAL VOC 2007+2012 is trained on augmented datasets of the fused maps.

C. Comparison With State-of-the-Art Methods

In this subsection, we evaluate the overall performance of the proposed method. The mainstream methods in the industry and our method are used for testing, and the results are compared. In a typical inspection system for defect detection, image acquisition is accomplished by a camera with appropriate illumination, and Faster-RCNN-r50, SSD300, YOLOV3 series object detection models are employed to detect the defects. On the other hand, the proposed method in this paper is adopted to collect images of the same scene, and our network is used for defect detection. The laptop manufacturer supplies the ground truth. In this experiment, Precision and Recall combined with F1 Score that considers both precision and recall are reported to evaluate the performance of the networks. Precision, recall, and F1 Score are defined as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

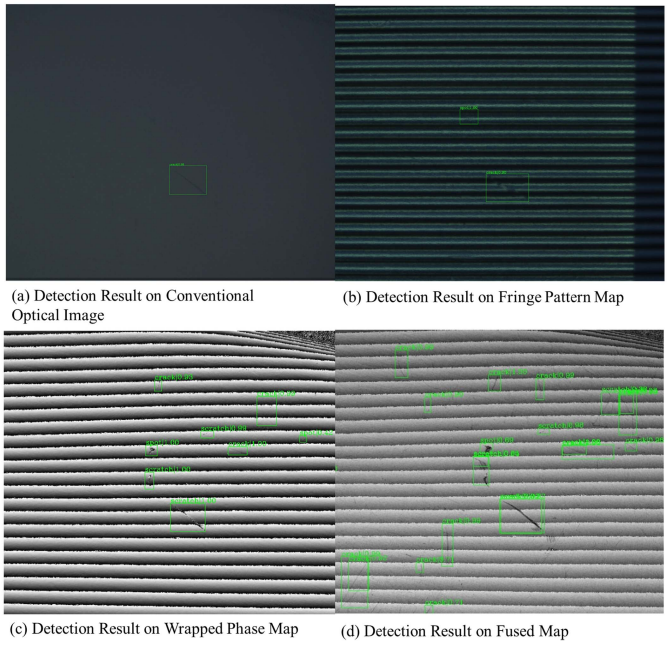
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

$$\text{F 1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

where TP is a positive example that is successfully predicted (Ture Positive), FP is a negative example that is misjudged as a positive example by the model (False Positive), and FN is a positive example that is falsely predicted as a negative example by the model (False Negative).

The results are reported in Table VII, which reveals that a suitable image acquisition method is also required to obtain satisfactory detection results in addition to an excellent algorithm. The mainstream methods' poor performance is not due to the algorithm's problem but due to the inadequate characteristics of the features.

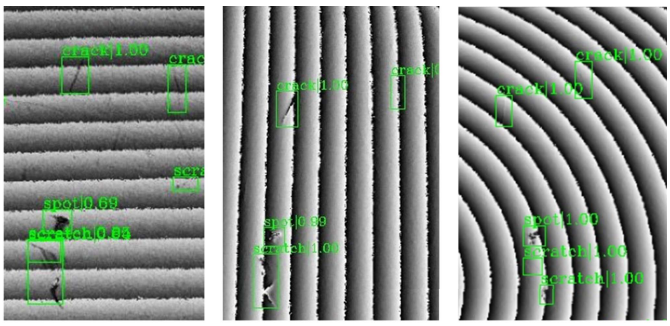
In contrast, a comprehensive description of defects in conjunction with a suitable inspection network enables our system to outperform the mainstream method's accuracy. The inspection results can be seen in Fig. 10. The recall and precision of various defects are higher than the mainstream methods currently used in production. At the same time, the inspection speed can also meet the inspection requirements for panel surface defects.



(a) Detection Result on Conventional Optical Image (b) Detection Result on Fringe Pattern Map

(c) Detection Result on Wrapped Phase Map (d) Detection Result on Fused Map

Fig. 10. Detection results of different image acquisition methods in the same scene (when using traditional lighting method (a), most of the defects are difficult to detect. Our method can find as many defects as possible.)



(a) Horizontal Line (b) Vertical Line (c) Circle

Fig. 11. Comparison results of the proposed method tested on different fringe shapes. No matter what stripes are used, defects can be identified.

We conducted the experiments by changing the fringe's shape and using the same method to detect defects on the laptop screens of the same batch of laptops, further verifying the robustness of our approach. The laptop screen and panel materials are somewhat different, but they also have the characteristics of a smooth surface. The types of defects can also be attributed to three common types of defects on smooth

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 07:17:48 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
THE PROPOSED METHOD TESTED ON DIFFERENT SMOOTH SURFACE AND FRINGE SHAPE

Reflective Surface Tpye	Fringe Shape	Scratch			Crack			Spot		
		Recall	Precision	F1 Score	Recall	Precision	F1 Score	Recall	Precision	F1 Score
laptop panel	horizontal line	0.997	0.939	0.967	0.993	0.958	0.975	0.961	0.907	0.933
	vertical lines	0.826	0.817	0.821	0.907	0.961	0.933	0.922	0.889	0.905
	circle	0.804	0.907	0.852	0.923	0.873	0.897	0.834	0.951	0.889
laptop screen	horizontal line	0.955	0.848	0.898	0.965	0.818	0.885	0.806	0.941	0.868
	vertical lines	0.838	0.841	0.839	0.966	0.942	0.954	0.846	0.841	0.843
	circle	0.842	0.862	0.852	0.807	0.918	0.859	0.802	0.924	0.859

surfaces. The results in Table VIII and Fig. 11 show that our method is minimally affected by fringe shape choice and performs equally well on other smooth surfaces. Our approach enables the defects can be identified more accurately, which is robust and strong in generalization.

V. CONCLUSION

This paper presented a new surface defect detection method to solve the lighting problem in the defect detection process. Given the lack of semantic features of defects, we proposed to utilize the wrapped phase maps obtained by the phase-shift method for object detection. In terms of algorithm, a more suitable defect detection backbone, the VoVecwNet, was proposed. Image fusion technology was adopted for data optimization to fuse the modulation map and the wrapped phase map. The two-source data complemented each other so that the defect characteristics were better represented. Experiments showed that our method achieved an accuracy of 97.5% and a reference time of 0.0762s on GTX 2080ti GPU in defect detection, which could meet the requirements of industrial manufacturing.

The method we proposed also has limitations. Although the deep learning method model has a high degree of uniformity, manual participation in the training process is negligible. It still needs to be manually labelled to define the defect category. Unknown defects on the surface will interfere with the detection process. In the future, we will continue to optimize the algorithm structure to solve these problems.

REFERENCES

[1] H. A. Abeysundara, H. Hamori, T. Matsui, and M. Sakawa, "A neural network approach for non-contact defect inspection of flat panel displays," *Procedia Comput. Sci.*, vol. 22, pp. 20–29, Jan. 2013.

[2] W. Ming *et al.*, "A comprehensive review of defect detection in 3C glass components," *Measurement*, vol. 158, Jul. 2020, Art. no. 107722.

[3] M. Chang, B.-C. Chen, J. L. Gabayno, and M.-F. Chen, "Development of an optical inspection platform for surface defect detection in touch panel glass," *Int. J. Optomechatron.*, vol. 10, no. 2, pp. 63–72, 2016.

[4] S. Konishi, A. L. Yuille, J. M. Coughlan, and S. C. Zhu, "Statistical edge detection: Learning and evaluating edge cues," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 1, pp. 57–74, Jan. 2003.

[5] H.-F. Ng, "Automatic thresholding for defect detection," *Pattern Recognit. Lett.*, vol. 27, no. 14, pp. 1644–1649, Oct. 2006.

[6] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[7] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2016, pp. 21–37.

[8] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*. [Online]. Available: http://arxiv.org/abs/1804.02767

[9] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*. [Online]. Available: http://arxiv.org/abs/2004.10934

[10] H.-W. Lei, B. Wang, H.-H. Wu, and A.-H. Wang, "Defect detection for polymeric polarizer based on faster R-CNN," *J. Inf. Hid. Multimedia Sign. Process*, vol. 9, pp. 1414–1420, Nov. 2018.

[11] X. Xu, H. Zheng, Z. Guo, X. Wu, and Z. Zheng, "SDD-CNN: Small data-driven convolution neural networks for subtle roller defect inspection," *Appl. Sci.*, vol. 9, no. 7, p. 1364, Mar. 2019.

[12] Z. Luo, X. Xiao, S. Ge, Q. Ye, S. Zhao, and X. Jin, "ScratchNet: Detecting the scratches on cellphone screen," in *Proc. Chin. Conf. Comput. Vis.* Berlin, Germany: Springer, 2017, pp. 178–186.

[13] L. Li, Z. Wang, F. Pei, and X. Wang, "Improved illumination for vision-based defect inspection of highly reflective metal surface," *Chin. Opt. Lett.*, vol. 11, no. 2, 2013, Art. no. 021102.

[14] E. N. Malamas, E. G. M. Petrakis, M. Zervakis, L. Petit, and J.-D. Legat, "A survey on industrial vision systems, applications and tools," *Image Vis. Comput.*, vol. 21, no. 2, pp. 171–188, Feb. 2003.

[15] D. Li, L.-Q. Liang, and W.-J. Zhang, "Defect inspection and extraction of the mobile phone cover glass based on the principal components analysis," *Int. J. Adv. Manuf. Technol.*, vol. 73, nos. 9–12, pp. 1605–1614, Aug. 2014.

[16] D. Martin, D. M. Guinea, M. C. García-Alegre, E. Villanueva, and D. Guinea, "Multi-modal defect detection of residual oxide scale on a cold stainless steel strip," *Mach. Vis. Appl.*, vol. 21, no. 5, pp. 653–666, Aug. 2010.

[17] J. Huang and Q. Wu, "A new reconstruction method based on fringe projection of three-dimensional measuring system," *Opt. Lasers Eng.*, vol. 52, pp. 115–122, Jan. 2014.

[18] S. S. Gorthi and P. Rastogi, "Fringe projection techniques: Whither we are?" *Opt. Lasers Eng.*, vol. 48, no. 2, pp. 133–140, Feb. 2010.

[19] X.-Y. Su, A. M. Zarubin, and G. von Bally, "Modulation analysis of phase-shifted holographic interferograms," *Opt. Commun.*, vol. 105, nos. 5–6, pp. 379–387, Feb. 1994.

[20] F. W. Y. Chan, "Reflective fringe pattern technique for subsurface crack detection," *NDT & E Int.*, vol. 41, no. 8, pp. 602–610, Dec. 2008.

[21] L. Huang, C. S. Ng, and A. K. Asundi, "Fast full-field out-of-plane deformation measurement using fringe reflectometry," *Opt. Lasers Eng.*, vol. 50, no. 4, pp. 529–533, Apr. 2012.

[22] H. Li and X.-J. Wu, "DenseFuse: A fusion approach to infrared and visible images," *IEEE Trans. Image Process.*, vol. 28, no. 5, pp. 2614–2623, May 2019.

[23] Y. Lee and J. Park, "CenterMask: Real-time anchor-free instance segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 13906–13915.

[24] J. Lin, Y. Yao, L. Ma, and Y. Wang, "Detection of a casting defect tracked by deep convolution neural network," *Int. J. Adv. Manuf. Technol.*, vol. 97, nos. 1–4, pp. 573–581, Jul. 2018.

[25] X. Tao *et al.*, "Wire defect recognition of spring-wire socket using multitask convolutional neural networks," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 8, no. 4, pp. 689–698, Apr. 2018.

[26] Y. Li, H. Huang, Q. Xie, L. Yao, and Q. Chen, "Research on a surface defect detection algorithm based on MobileNet-SSD," *Appl. Sci.*, vol. 8, no. 9, p. 1678, Sep. 2018.

[27] Y.-F. Chen, F.-S. Yang, E. Su, and C.-C. Ho, "Automatic defect detection system based on deep convolutional neural networks," in *Proc. Int. Conf. Eng., Sci., Ind. Appl.*, Aug. 2019, pp. 1–4.

[28] Y. Huang, C. Qiu, and K. Yuan, "Surface defect saliency of magnetic tile," *Vis. Comput.*, vol. 36, no. 1, pp. 85–96, Jan. 2020.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 07:17:48 UTC from IEEE Xplore. Restrictions apply.

[29] T.-Y. Lin *et al.*, "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* Berlin, Germany: Springer, 2014 pp. 740–755.
[30] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 4700–4708.
[31] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
[32] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[33] I. Cosmin Duta, L. Liu, F. Zhu, and L. Shao, "Improved residual networks for image and video recognition," 2020, *arXiv:2004.04989*. [Online]. Available: http://arxiv.org/abs/2004.04989
[34] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.
[35] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.
[36] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Oct. 2017, pp. 2961–2969.
[37] K. Song and Y. Yan, "A noise robust method based on completed local binary patterns for hot-rolled steel strip surface defects," *Appl. Surf. Sci.*, vol. 285, pp. 858–864, Nov. 2013.
[38] M. Everingham, L. Van Gool, C. K. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes challenge 2007 (VOC2007) results," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338.
[39] K. He, R. Girshick, and P. Dollár, "Rethinking ImageNet pre-training," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2019, pp. 4918–4927.
[40] G. Finlayson, S. Hordley, G. Schaefer, and G. Y. Tian, "Illuminant and device invariant colour using histogram equalisation," *Pattern Recognit.*, vol. 38, no. 2, pp. 179–190, Feb. 2005.
[41] M. Lecca and S. Messelodi, "Linking the von kries model to Wien's law for the estimation of an illuminant invariant image," *Pattern Recognit. Lett.*, vol. 32, no. 15, pp. 2086–2096, Nov. 2011.



Jiamin Tao received the B.E. degree from the School of Electronic and Information Engineering, Beijing Jiaotong University, China, in 2019. She is currently pursuing the degree with the School of Information and Optoelectronic Science and Engineering, South China Normal University.



Yongjian Zhu received the Ph.D. degree from the Shanghai Institute of Optics and Fine Mechanics, Chinese Academy of Sciences in 2007. His research interests include machine vision and default detection based on deep learning.



Wenyi Liu received the B.E. degree from the Bejing University of Posts and Telecommunications in 2019. She is currently pursuing the master's degree with the School of Information and Optoelectronic Science and Engineering, South China Normal University. Her current research focuses on deep learning and optical communication.



Frank Jiang received the Ph.D. degree from the University of Technology Sydney in 2008. He has three and half years of postdoctoral research experience at New South Wales (UNSW). His main research interests include the IoT security, biologically inspired learning schemes and their applications in the context-aware systems, and blockchain techniques.



Hongzhan Liu received the Ph.D. degree from the Chinese Academy of Sciences. He is a Professor with the School of Information and Optoelectronic Science and Engineering, South China Normal University. He is a member of the Guangdong Provincial Key Laboratory of Nanophotonic Functional Materials and Devices. His research interests include satellite-ground laser communication, precision measurement, sensing, and the Internet of Things technology.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on October 14,2026 at 07:17:48 UTC from IEEE Xplore. Restrictions apply.